

called on. At the height of the fighting, the US Air Force, Navy, and Marines were conducting an average of 800 tactical air sorties a day. Aging piston-engined aircraft, especially A-1 Skyraiders, were notably successful in the ground-attack role, their slower speed helpful in picking out targets, but the bulk of close air-support missions were flown by high-performance jets such as F-4 Phantom IIs and F-100 Super Sabres.

Heavy bombing

The B-52s constituted the most powerful attack force used in South Vietnam. Until the very end of America's involvement in the conflict, these quintessential strategic bombers were barred from operating over North Vietnam, denying them their obvious role. They spent most of the war carpet-bombing areas of paddy field and jungle, seeking to destroy communist supply routes and base camps as well as troop formations. Extraordinary technological ingenuity was devoted to finding targets for the B-52s. Aircraft scattered listening devices and other sensors across the hills and forests where the famed Ho Chi Minh Trail brought men and supplies from North Vietnam into the South. When troop or transport movements were detected, flights of B-52s would fly from their base on Guam in the Pacific, and release their bombloads, devastating several square miles of ground at a time.

The lack of proportion between the air power deployed and its results was nowhere greater than in the campaign against these supply routes. It is estimated that a larger tonnage of bombs was dropped on the Trail than had been dropped on all fronts during World War II. Although estimates are speculative, it may have, on average, taken 100 tons of bombs to kill a single communist soldier.

Paradoxically, the firepower that the Americans had at their disposal in South Vietnam was in fact excessive. When operating in remote unpopulated zones, the scale of the devastation US aircraft caused was no real matter for concern. But when fighting moved into populated rural areas, the "collateral damage" inflicted on local noncombatants was a grave embarrassment to the United States, weakening support for the war at home. The use of napalm, which had first been employed in World War II, was a particular focus of controversy and protest. To the US military it was simply a very effective weapon against troops dug in to tunnels or trenches. But a photograph of a young Vietnamese girl burnt by napalm became the single most famous image of the war.

Bell AH-1 Cobra



THE IDEA OF AN ATTACK HELICOPTER emerged in the early 1960s when armies realized how vulnerable their troop-carrying helicopters were to ground fire, especially during counterinsurgency operations. Since it was difficult to fly fixed-wing aircraft with helicopters during a mission, it was decided that the best helicopter escort would be another helicopter.

At the start of the Vietnam War, the Americans used the UH-1 Iroquois ("Huey") helicopter to suppress ground fire. This was simply a more heavily armed version of a troop-transporter.

In a sense, the AH-1 Cobra represented only a limited step forward: about 85 percent of its components were identical to those found in the UH-1. It was, however, the first rotating-wing aircraft specifically designed as a gunship, and, as such, was a milestone in the development of military helicopters.

First flown in 1965, AH-1s arrived in Vietnam two years later. In addition to "escorting" duties,

the AH-1s were also sent on search-and-destroy missions, working in tandem with OH-6 Cayuse scout helicopters in what was termed a "Pink Team". Shadowed by a Cobra at about 1,500ft (500m), the low-flying scout helicopters would seek out the Viet Cong. When the enemy opened fire on the OH-6, the Cobra would fly down and lay waste to the area from where the fire was coming. After the Vietnam War, the AH-1 was adapted for a new role as a tank-buster. It was equipped with guided antiarmor missiles

and an upgraded engine to give it more chance of survival on a conventional battlefield (where it would have to contend with ground fire, hostile aircraft, and missiles). Equipped with a multitude of advanced avionics – such as fire-control computers and infrared receivers for night fighting – by the 1980s the Cobra had evolved into an extremely sophisticated fighting machine.

